



SHOPEZ PLATFORM SUITE

REQUIREMENT ANALYSIS

Requirement Analysis Document (RAD)

Functional Specs, Performance Limits & Traceability Matrices

PROJECT NAME:	ShopEZ Premium E-Commerce Platform
REPOSITORY NAME:	VIP-C2-ShopEZ-E-commerce-Application
PREPARED BY:	P Pranay
DATE:	June 2026
VERSION / STATUS:	v1.0.0 / Final Approved

1. Core Functional Requirements

The features implemented in ShopEZ represent complex, production-grade logic partitioned into primary epics:

Epic A: Core Catalog & Merchandising

- FR-1: Real-time title matching query for client autocomplete inputs.
- FR-2: Dynamic category grids navigating user straight to specific collection links.
- FR-3: Item variations (sizes, color, storage) rendering specific stock check checks.

Epic B: Secure Checkouts & Wallet Systems

- FR-4: Coupon calculations logic deducting flat or percentage savings (e.g. coupon Welcome10).
- FR-5: Customer sandbox wallet subtracting balance on checkout submit.
- FR-6: Timeline steps displaying Processing -> Packed -> Shipped -> Out for Delivery -> Delivered.

2. Non-Functional Performance Limits

- Security: User token blacklisting via blocked middleware inside Express server routes.
- Availability: Failover response from local JSON files when connection to main MongoDB fails.
- Portability: Responsive, device-agnostic interface styled via vanilla CSS parameters.

REQUIREMENT VERIFICATION MAPPING MATRIX

Req ID	Feature Name	UAT Test Scenario	Status
FR-1	Search Autocomplete	Enter text in navbar search and assert suggests	Verified
FR-3	Product Variations	Choose Size/Color and assert cart item contains choice	Verified
FR-4	Coupon Deductions	Apply FLASHSALE30 and assert 30% savings calculated	Verified
FR-5	Sandbox Wallet Pay	Assert user wallet balance decreases by order total	Verified

